

ARAB YOUTH SPACE HACKATHON · CHALLENGE 813 · TEAM T0006 · PALESTINE

Theme: Ecosystem Health, Biodiversity & Blue Carbon

Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves — from alert to evidence.

91 stands

monitored, Abu Dhabi + Saudi Tarut Bay

5

real conversions, confirmed at sub-metre

≈1

false alarm per stand-year, calibrated

THE PROBLEM

Restoration is scaling faster than monitoring

100 million mangroves by 2030

UAE national pledge, COP26 2021 — Abu Dhabi Mangrove Initiative launched 2022

100+ million mangroves by 2030

Saudi Arabia, Saudi Green Initiative

The regulator is asking for monitoring

EAD + IUCN Mangrove Specialist Group: first mangrove monitoring guide for the GCC, Feb 2026

SDG 13 · 14 · 15

Climate action · Life below water · Life on land

Today: periodic field surveys and one-off studies — a 2025 regional review calls Gulf MRV “uneven”. No standing alert layer for individual stands. No independent stock check for carbon claims.

BUSINESS USE CASE

Who uses it, and for which decision

Environment agencies

DECISION

Which stands to inspect this month? Is a protected stand intact?

TODAY, WITHOUT US

Periodic field surveys and one-off studies

Planting & carbon-project developers

DECISION

Is the carbon stock we report still there?
Evidence for verification.

TODAY, WITHOUT US

Field plots and drone surveys, campaign by campaign

Coastal developers & regulators

DECISION

Did construction stay outside the mangrove line?

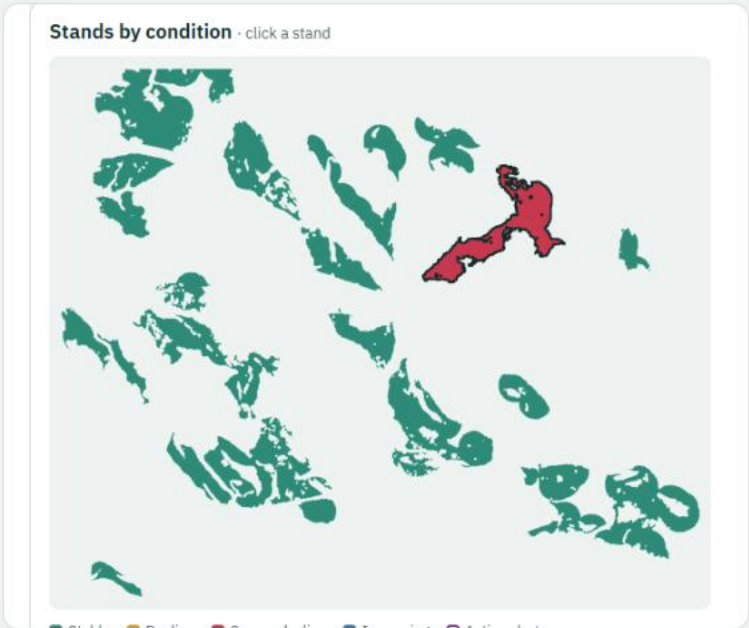
TODAY, WITHOUT US

Comparing two dated images by eye

What they get: a monthly alert per stand *and* per 200 m cell, a map of where to look, the carbon at stake, and a site report — at about one false alarm per stand per year.

DATA & STUDY AREA

25 mangrove stands near Abu Dhabi



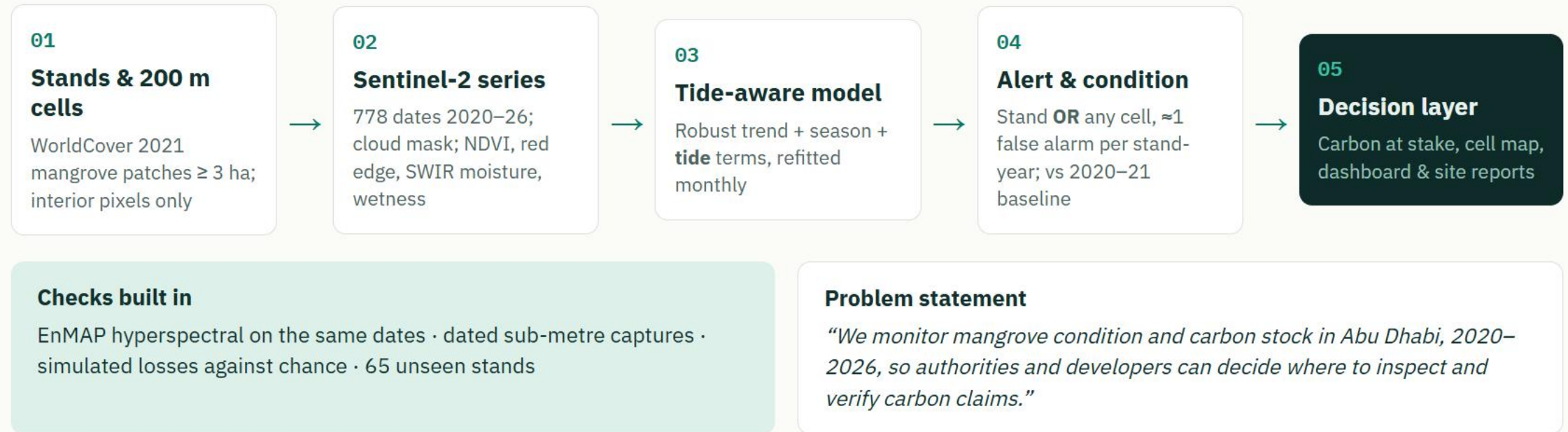
25 patches ≥ 3 ha (ESA WorldCover 2021) in the densest 13×10 km block near Abu Dhabi. Plus 66 unseen stands (Abu Dhabi, Saudi Tarut Bay) to test transfer.

Source	Role	Access
Sentinel-2 (ESA/Copernicus)	Time series 2020–26, 5-day revisit	Free — 778 dates
ESA WorldCover 2021	Stand outlines	Free (CC BY 4.0)
EnMAP (DLR)	Independent check, 224 bands	Free — 2 scenes
Field carbon plots	Density, 24 plots / 4 sites	CC0, Schile 2016
300 reference points	Extent check	AI-assisted, blind

No source in this PoC depends on an allocation we do not already have.

APPROACH

From satellite pixels to an inspection decision

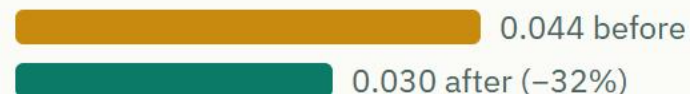


The tide looks like tree loss — until you correct for it

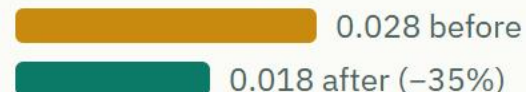
Mangrove canopies flood on every tidal cycle. One satellite pass mixes canopy signal with the water and mud underneath. On our stands, a wetness index alone explains most of the scene-to-scene noise — correlation about -0.9 .

Our model regresses each stand's index on season, trend and tidal state before it ever flags an anomaly.

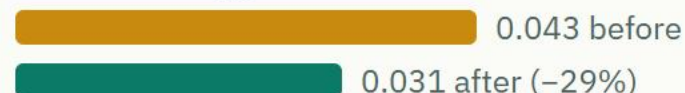
NDVI



NDRE (red edge)

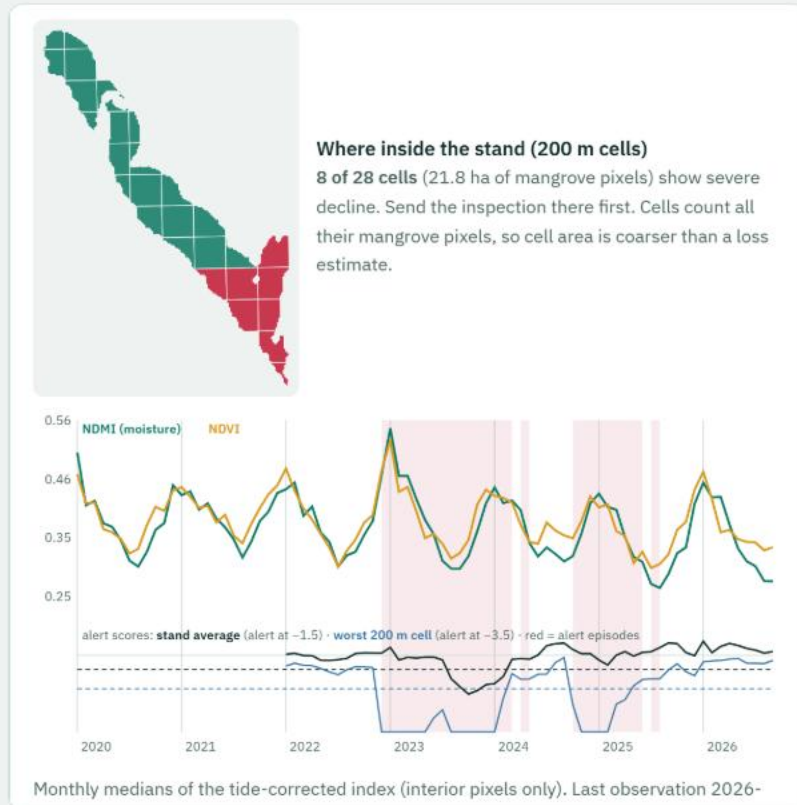


NDMI (canopy moisture)



Tide correction cuts date-to-date noise by roughly 30–35% on every index, before any anomaly model runs.

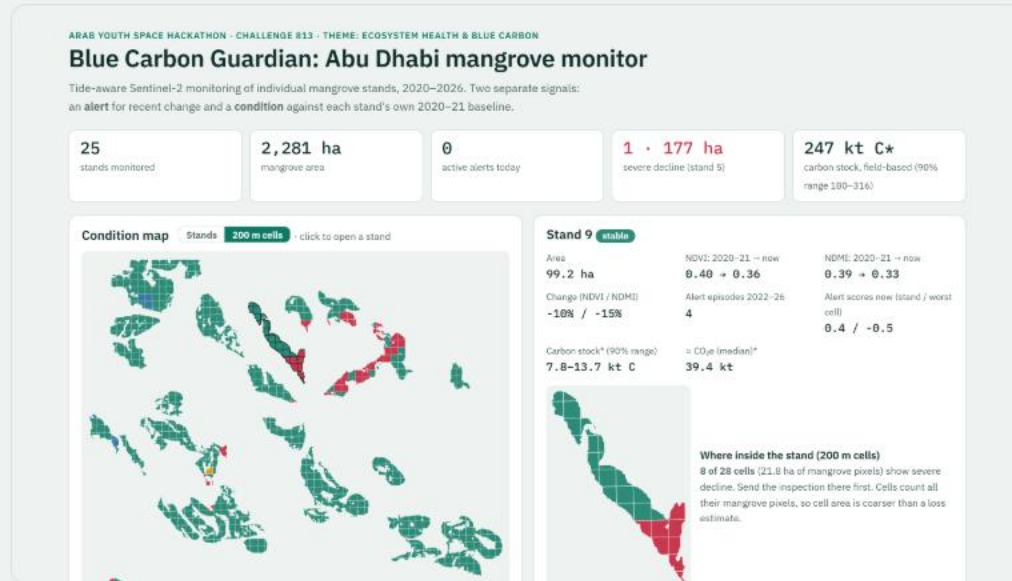
A stand average hides partial loss. 200 m cells do not.



- ✓ Stand 9 reads “stable” on average — but **8 of its 28 cells** are in severe decline: exactly the part that converted.
- ✓ Three of our five real conversions (stands 12, 18 and 6) were found **only** by the cells.
- ✓ One alert rule: **stand average OR any cell**, calibrated together to about one false alarm per stand-year (0.96 measured).
- ✓ A partial loss is caught within 60 days in **73%** of simulated cases with cells, **25%** with the stand average alone — 13% by chance.

PRODUCT

From signal to decision

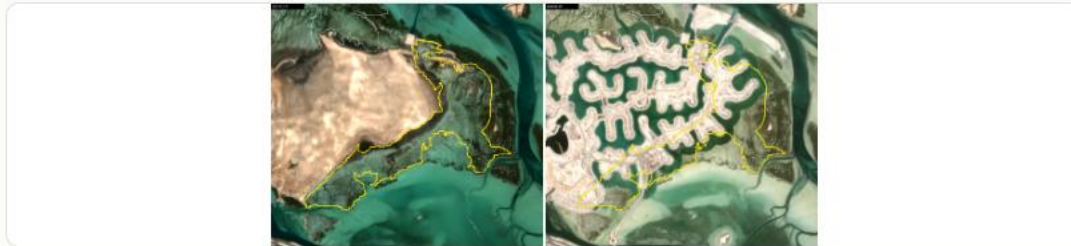


- ✓ One alert: stand average OR any 200 m cell, about one false alarm per stand-year
- ✓ Where inside the stand: a 200 m cell map for the inspector
- ✓ Condition vs the stand's own 2020–21 baseline, over a full year; carbon with a range
- ✓ One printable site report per stand, ready for a compliance file
- ✓ Templated wording — every sentence traces back to a number, never invented

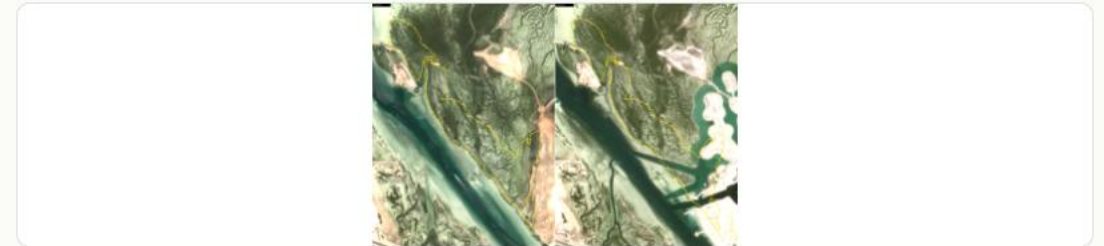
Built for: environment agencies · mangrove & carbon-project developers · EIA consultants & regulators

VALIDATION 2 – REAL EVENTS

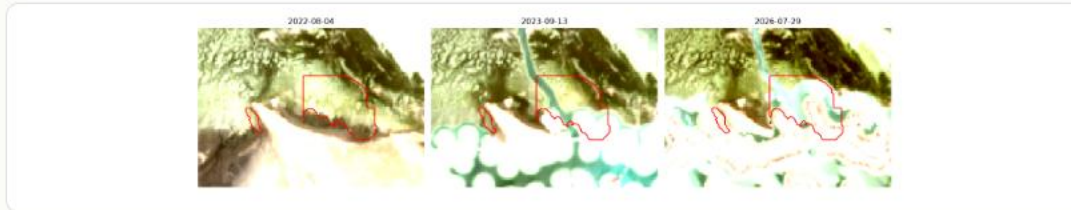
Five real conversions, confirmed on sub-metre imagery



Stand 5, 2021→2026: a reclamation with finger canals. Our alert on from Oct 2022, mostly from the cells that later converted.



Stand 9, 2020→2026: a canal-style development takes the south-east. Stand average “stable”; 8 of 28 cells in severe decline.



Stand 12, 2022→2023→2026: a new canal and reclamation take the southern edge. Found only by the cells; alert from Dec 2022.

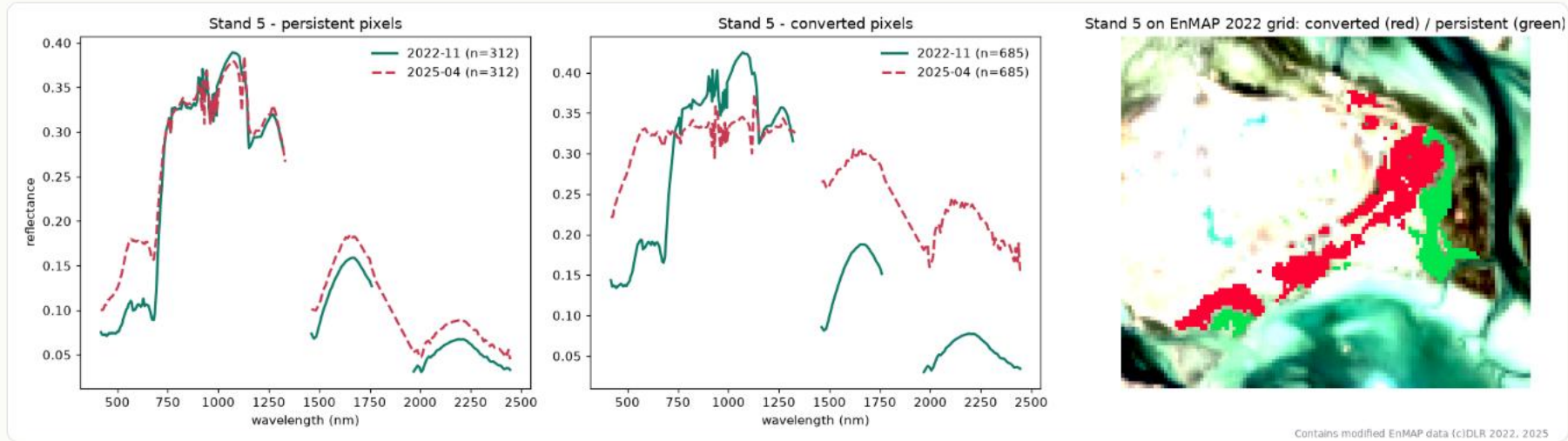


Stand 18, 2020→2023→2026: the same development reaches it in 2025. Found only by the cells.

Dated 0.3–0.5 m captures (WorldView-2/3, Legion-1) confirm all four shown — and a fifth, stand 6, where a lagoon replaced part of the stand. At every site our alert began between the last undisturbed capture and the first one showing works — detection of works as they happen, not advance warning. All are partial losses: carbon figures are upper bounds.

VALIDATION 3 – HYPERSENSPECTRAL (ENMAP)

An independent check, and a material fingerprint



$r = 0.84-0.99$

EnMAP vs our Sentinel-2 indices, same date,
across 25 and 14 stands

62 vs 26 ha

Stand 5 conversion: EnMAP pixels vs our
conservative Sentinel-2 flag

F1 0.74 vs 0.72

Full spectrum vs Sentinel-2-like bands as a
classifier: no significant gain, and we say so

VALIDATION 1 – SIMULATION

Can we detect a loss? Tested against pure chance

We injected artificial canopy loss into the real 2020–2026 series of stands with no known change, holding false alarms to about one per stand per year. A detection counts only if a **new** alert starts after the loss begins. The same test with **no** loss gives the chance rate.

Simulated loss	Alert	Detected	By chance
A tenth of a stand loses half its canopy	within 60 days	73%	13%
A whole stand loses 10% of its canopy	within 6 months	63%	36%
A whole stand loses 20% of its canopy	within 6 months	83%	36%

One alert rule for the whole product: the **stand average OR any 200 m cell**. The cells catch partial losses — how all four real conversions happened — that a stand average dilutes.

We corrected our own earlier test: it counted alerts that were already running, and it had no chance baseline (we had reported 58% / 90%). About 160–240 trials per row, so ± 7 points. Out of sample (65 unseen stands, thresholds fixed): 0.79 false-alarm episodes per stand-year vs 0.96 calibrated.

LIMITATIONS

Where it breaks — and what we do not claim

Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.

10 m pixels

Patches under ~0.6 ha and young plantings are missed — Bahrain's small stands included.

Imagery, not site visits

Five conversions confirmed on dated sub-metre captures; not yet on permits or field records.

Simulated losses

Canopy-cover loss, not leaf stress; every detection rate shown next to its chance rate; ± 7 points.

Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).

Carbon

Area $\times$ field density from four sites (2013–14): a stock with a range, never a sequestration rate.

We withdrew our own first backtest figure (58% / 90%) when a null test showed its metric counted alerts already running.

CARBON

Carbon stock, grounded in field measurements

We use an open field dataset (Schile et al. 2016, CC0): 24 plots at four sites near our stands, combining tree carbon with soil organic carbon to 1 metre depth.

108 t C/ha

local mean density (90% CI 79–138)

247 kt C

region stock (90% range 180–316)

907 kt CO₂e

region equivalent, median

Stand	Area	Carbon (median)	Note
5	176.5 ha	19.1 kt C	Severe decline — upper bound, includes reclaimed area
9	99.2 ha	10.8 kt C	Stable on average — 8 of 28 cells in severe decline

We tested whether Sentinel-2 indices predict field-measured tree carbon (95 plots, 16 sites). The link is weak (best: red edge, $p=0.38$) — too weak for a biomass model. So carbon here is area × field density, not imagery-derived biomass.

BUSINESS MODEL

A monitoring service with honest unit economics

Customers

Environment agencies (EAD, MOCCA, Saudi NCVC)

Planting programmes: ADNOC (10 M), Red Sea Global (50 M), carbon developers

Coastal developers & EIA consultants

Offer · price (hypothesis)

Monitor: USD 12k per site per year ($\approx$ USD 5 / ha)

Evidence: + high-res check of every alert, audit pack: USD 25k

Development watch: USD 8k per project per year

Cost to serve $\approx$ USD 3.3k / site / yr

Compute < USD 5 (**measured:** full 2020–26 archive in under 75 min)

High-res images only when an alert fires: USD 500–750 (vs 5–8k for monthly coverage)

Analyst time (assumption): USD 2k

Competition. Global Mangrove Watch offers free monthly loss alerts (Africa first, expanding; Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell level, with carbon and reports — and can use GMW extent as input.

THE HONEST MARKET

Gulf + Red Sea mangroves $\approx$ 16 sites $\rightarrow$ USD 0.2–0.4 M / yr: a niche. The real potential: a national MRV contract for a 100-million-mangrove programme, a monitoring layer on gIQ, licensing.

Prices and analyst time are hypotheses for the pilot to test. Scenario and all 13 sources: docs/business_plan_draft.md.

ROADMAP

What is next

NOW — POC

Stand + 200 m cell monitor, dashboard & site reports, field-grounded carbon, 5 real conversions confirmed, EnMAP check. Unchanged on 65 unseen stands (incl. Saudi Tarut Bay): 0.79 false alarms per stand-year vs 0.96 calibrated

IN PROGRESS

More EnMAP dates per season, to separate seasonal and tidal change from true leaf-level stress

INCUBATION

Satellite 813, gIQ deployment, calibrated cell-level alerts, scale to Saudi Arabia & Bahrain, a pilot customer

Two limits we are designing around, on purpose

- Bahrain's own stands are too small for 10–30 m pixels — a case for very-high-resolution data at incubation.
- Our five events are confirmed on dated sub-metre imagery, not yet on site records — matching them to permits with a partner is the first pilot task.

TEAM T0006

Built end to end by one engineer

Nahla Nabil

Bahrain — idea, satellite data pipeline, tide-aware model, validation, dashboard, business case. AI/LLM engineer; NASA Space Apps local lead.

What we are asking for

Mentor feedback on hyperspectral interpretation of Gulf mangroves & sabkha vegetation — and a path into incubation to validate with a real restoration partner.

Thank you. — Blue Carbon Guardian, Team T0006